

Math 67 – HW 1 – Solutions of graded problems

Modified: 11:53am, 2/1/2010

Chapter 1: Proof problem 1

Here is a different proof than the one I started in discussion.

Claim. Let a, b, c , and d be real numbers, and consider the system

$$ax_1 + bx_2 = 0$$

$$cx_1 + dx_2 = 0.$$

If $ad - bc \neq 0$, then $x_1 = x_2 = 0$.

Proof. Suppose $x_1 \neq 0$. Then we can divide by x_1 in (1) and (2) to get

$$a + b \frac{x_2}{x_1} = 0 \quad \text{and} \quad c + d \frac{x_2}{x_1} = 0.$$

Then

$$\frac{x_2}{x_1} = -\frac{a}{b} = -\frac{c}{d},$$

so $ad = bc$. But this is not possible because $ad - bc \neq 0$. Then we must have $x_1 = 0$.

Similarly, suppose $x_2 \neq 0$. Then we can divide (1) and (2) by x_2 , yielding

$$a \frac{x_1}{x_2} + b = 0 \quad \text{and} \quad c \frac{x_1}{x_2} + d = 0.$$

Then

$$\frac{x_1}{x_2} = -\frac{b}{a} = -\frac{d}{c},$$

so $bc = ad$. But this cannot be so, because $ad - bc \neq 0$. So $x_2 = 0$.

We just showed that $x_1 = 0$ and $x_2 = 0$.

Chapter 2: calculational

3. We want $r > 0$ and $\theta \in [0, 2\pi)$ such that $(1 - i)/\sqrt{2} = re^{i\theta}$.

$$r = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)} = 1.$$

There are several ways to express θ but we want $\theta \in [0, 2\pi)$.

$$\theta = -45^\circ = -\pi/4 = 7\pi/4.$$

Then the polar form of the given complex number is $e^{7\pi i/4}$.

4(d) The solution(s) $z \in \mathbb{C}$ for the equation $z^3 - 4i = 0$.

$$z^3 = 4i = 4e^{\frac{\pi}{2}i}.$$

The rest follows directly from Theorem 2.3.3. Solutions are

$$z_k = 4^{1/3} e^{i/3(\frac{\pi}{2} + 2\pi k)} \quad \text{for } k = 0, 1, 2.$$

Alternatively stated, the set of solutions is

$$z = \left\{ 4^{1/3} e^{\frac{\pi}{6}i}, 4^{1/3} e^{\frac{5\pi}{6}i}, 4^{1/3} e^{\frac{3\pi}{2}i} \right\}.$$

(1)

(2)

5(c) It follows from Theorem 2.2.12 that

$$\left| \frac{i(2 + 3i)(5 - 2i)}{-2 - i} \right| = \frac{|i||2 + 3i||5 - 2i|}{|-2 - i|} = \frac{1 \cdot \sqrt{13} \cdot \sqrt{29}}{\sqrt{5}} = \sqrt{\frac{377}{5}}$$

Chapter 2 proof problem 3

Claim. Let $w, z \in \mathbb{C}$. Then $|z - w|^2 + |z + w|^2 = 2(|z|^2 + |w|^2)$.

Proof. Since $z, w \in \mathbb{C}$, there are real numbers $z_1, z_2, w_1, w_2 \in \mathbb{R}$ such that $z = z_1 + z_2i$, and $w = w_1 + w_2i$. Then

$$\begin{aligned} |z - w|^2 + |z + w|^2 &= |(z_1 + z_2i) - (w_1 + w_2i)|^2 + |(z_1 + z_2i) + (w_1 + w_2i)|^2 \\ &= |(z_1 - w_1) + (z_2 - w_2)i|^2 + |(z_1 + w_1) + (z_2 + w_2)i|^2 \\ &= (z_1 - w_1)^2 + (z_2 - w_2)^2 + (z_1 + w_1)^2 + (z_2 + w_2)^2 \\ &= (z_1^2 - 2z_1w_1 + w_1^2) + (z_2^2 - 2z_2w_2 + w_2^2) + (z_1^2 + 2z_1w_1 + w_1^2) + (z_2^2 + 2z_2w_2 + w_2^2) \\ &= 2(z_1^2 + z_2^2 + w_1^2 + w_2^2) \\ &= 2(|z|^2 + |w|^2) \end{aligned}$$

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